# LOMBA KOMPETENSI SISWA
# SMK TINGKAT NASIONAL KE-29
# TAHUN TAHUN 2021



## PRE-TEST PROJECT
## MODUL A – DAY 1 CLIENT SERVER

BIDANG LOMBA

**TEKNOLOGI INFORMASI SISTEM ADMINISTRASI JARINGAN**

***IT NETWORK SYSTEMS ADMINISTRATION***

## DESCRIPTION OF PROJECT

A small startup company's production environment contains numerous services within multiple Operating Systems. We will ask you to install, configure, and integrate the different services. You will have two work days to complete two different sets of tasks. Each set of tasks is to be completed within one work day. For the second work day, you will continue the work of another engineer, who should have fully configured systems according to the day one set of tasks. You may check, change, or update the existing systems, but we will only score the tasks you're supposed to do on the second work day.

# DAY 1 - NETWORK INFRASTRUCTURE

**Basic Configuration**

- Configure IP Address of all network devices according to the addressing table.
- Create SSH user 'patah' with password specified in the appendix.
- Allow server admins to SSH to all network devices.
- Restrict non-server PCS to SSH to all network devices.

**NAT and Routing**

- Configure Dynamic Routing for public addresses using Border Gateway Protocol
- Enable port NAT to all private addresses in sabang.net, indonesia.com, and merauke.net so that every device can connect to each other.
- Allow sabang.net to access merauke.net via encrypted VPN tunnel.
- On RTB-1, Create static NAT with Public IP 30.9.65.10 for HYPERV2 (172.17.11.102)

# DAY 1 - LINUX SERVICES

**Basic Configuration**

- Configure IP Address of all linux devices according to the addressing table.
- Create SSH user 'patah' with password specified in the appendix.
- Enable SSH access from any devices.

**Connectivity**

- Configure port NAT in fw.sabang.net to enable access to other networks.

**Sabang DNS**

- Access srv1 and srv2 and install bind9. Configure it to serve domain sabang.net and indonesia.com using the private addresses.
- Create A records of srv1.sabang.net, srv2.sabang.net, srvind.indonesia.com that points to their respective addresses.
- Create subdomain ns1.sabang.net that points to address of srv1.sabang.net
- Create subdomain ns2.sabang.net that points to address of srv2.sabang.net
- On both nameservers, create the following subdomains using address record:
    - www.sabang.net that points to the address of fw.sabang.net
    - mail.indonesia.com that points to the address of srvind.indonesia.com.
    - 100 user subdomain that points to the address of fw.sabang.net:
        - user001.public.sabang.net
        - user002.public.sabang.net
        - user003.public.sabang.net
        - …
        - user099.public.sabang.net
        - user100.public.sabang.net

**Sabang Web Hosting**

- Access srv1 and srv2 and install apache2. Configure it to serve all websites in sabang.net.
- Create /var/www/index.html default page with content specified in Appendix.
- Create /var/www/userXXX/index.html default page for 100 users with content specified in Appendix. Example:
    - /var/www/user001/index.html
    - /var/www/user002/index.html
    - …
    - /var/www/user099/index.html
    - /var/www/user100/index.html

- Create virtual host www.sabang.net serving /var/www/index.html
- Create 100 virtual hosts userXXX.public.sabang.net serving /var/www/userXXX/index.html . Example:
    - user001.public.sabang.net serving /var/www/user001/index.html
    - user002.public.sabang.net serving /var/www/user002/index.html
    - …
    - user099.public.sabang.net serving /var/www/user099/index.html
    - user100.public.sabang.net serving /var/www/user100/index.html

**Sabang Load Balancer**

- Access fw.sabang.net and install haproxy. Configure it to loadbalance web requests.
- Load Balance www.sabang.net with round robin algorithm.
- Load Balance 100 user websites with source ip algorithm.
    - user001.public.sabang.net
    - user002.public.sabang.net
    - user003.public.sabang.net
    - …
    - user099.public.sabang.net
    - user100.public.sabang.net

**Sabang Failover**

- Access srv1 and srv2 and install keepalived. Configure it to serve as a failover when one of the servers is down.
    - Use virtual IP 172.16.1.205
    - Use VRRP ID 205
    - Use script to check DNS service. When DNS service is down, move virtual IP to another server.

**Sabang iSCSI**

- Access srv1 and srv2 and install tgt. Configure it to serve iSCSI target in sabang.net.
    - Use device /dev/sdb, /dev/sdc, /dev/sdd, /dev/sde, /dev/sdf, /dev/sdg, /dev/sdh, /dev/sdi, /dev/sdj, /dev/sdk
    - A total of 20 disks will become iSCSI targets
- Access fw.sabang.net and install open-iscsi. Configure it to connect to iSCSI target in srv1 and srv2.
    - Don't create any filesystem on the disk.
    - Make sure both disks from srv1 and srv2 are available on fw.sabang.net.

**Sabang DHCP**

- Access fw.sabang.net and install any dhcp server. Configure to serve DHCP in sabang.net.
    - Network : 172.16.0.0/20
    - Reserved Addresses : 172.16.0.1 - 172.16.14.254
    - Gateway : 172.16.15.254

# DAY 1 - WINDOWS SERVICES

**Basic Configuration**

- Configure IP Address of all windows servers according to the addressing table.
- Determine and set gateway IP address to all windows servers.
- Configure hostname of all windows servers.
- Make all windows servers pingable from any devices.

**Active Directory and DNS**

- Access DC and Configure AD for domain merauke.net
- Create a normal AD user ‘patah’ with password specified in the appendix.
- Create DNS Address Record for servers in merauke.net according to their hostnames, for example DC.merauke.net
- Create DNS Record www.merauke.net pointing to IP Address of HYPERV1
- Create 25 DNS Records for users website pointing to IP Address of HYPERV1 and HYPERV2:
    - user01.merauke.net
    - user02.merauke.net
    - user03.merauke.net
    - …
    - user25.merauke.net

**Certificate Authority**

- Access DC and Configure CA to issue required Certificates by Linux Services
    - Common Name : LKSN2021-CA
- Generate certificates required by other services with the following DNS name:
    - www.sabang.net
    - *.public.sabang.net
    - www.merauke.net
    - *.merauke.net
    - Save these certificates and their private keys as one pfx file in C:\cert\
    - Use **Skills39** as export password
    - Use the DNS name as filename :
        - www.sabang.net.pfx
        - public.sabang.net.pfx
        - www.merauke.net.pfx
        - wildcard.merauke.net.pfx

# APPENDIX

**Users and Passwords**

| Username | Password |
|---|---|
| patah | KesultananDemak2021 |

**Network Devices Addressing Table**

| Device Name | IP Address CIDR | Network |
|---|---|---|
| RTA | 172.17.1.2/30 | sabang.net |
| | 192.168.199.1/21 | indonesia.com |
| | 30.9.65.1/30 | Public |
| ISP | 30.9.65.2/30 | Public |
| | 30.9.19.1/24 | Public |
| | 30.9.65.9/29 | Public |
| RTB-1 | 30.9.65.11/29 | Public |
| | 172.17.9.1/22 | merauke.net |
| RTB-2 | 30.9.65.12/29 | Public |
| | 172.17.9.2/22 | merauke.net |

**End Devices Addressing Table**

| Device Name | IP Address CIDR | Network |
|---|---|---|
| srv1.sabang.net | 172.16.1.201/20 | sabang.net |
| srv2.sabang.net | 172.16.1.202/20 | sabang.net |
| ani-pc | 172.16.1.101/20 | sabang.net |
| fw.sabang.net | 172.16.15.254/20 | sabang.net |

| | | |
|---|---|---|
| | 172.17.1.1/30 | sabang.net |
| srvind.indonesia.com | 192.168.192.168/21 | srvind.indonesia.com |
| DC | 172.17.10.100/22 | merauke.net |
| HYPERV1 | 172.17.11.101/22 | merauke.net |
| HYPERV2 | 172.17.11.102/22 | merauke.net |
| budi-pc | 30.9.19.10/24 | Public |

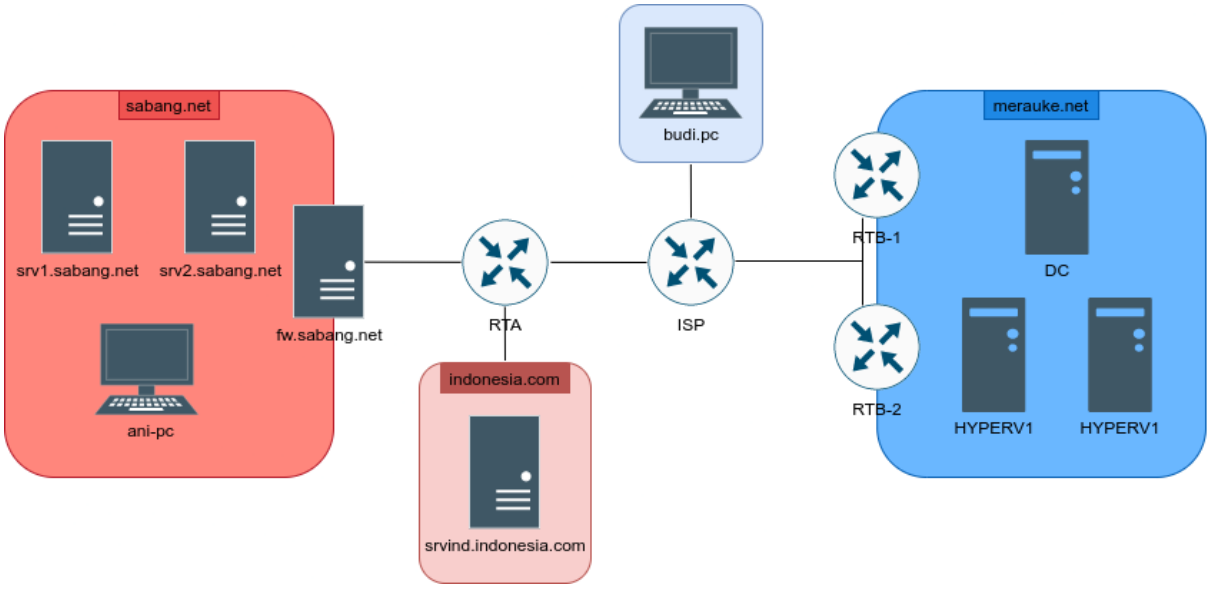
/var/www/index.html

```
<h1> Default Page </h1><br>
<br>
This page has not been modified by the owners.
```

/var/www/userXXX/index.html
[Replace XXX in the folder name and file content with user number, for example /var/www/user009/index.html]

```
<h1> Default Page for userXXX </h1><br>
<br>
This page has not been modified by the user.
```

# TOPOLOGY


sabang.net
srv1.sabang.net
srv2.sabang.net
fw.sabang.net
ani-pc
RTA
indonesia.com
srvind.indonesia.com
budi.pc
ISP
RTB-1
RTB-2
merauke.net
DC
HYPERV1
HYPERV1